Joseph Rich

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COMPUTATIONAL BIOLOGY • MEDICAL IMAGING • ARTIFICIAL INTELLIGENCE • APPLIED MATHEMATICS

Computational biology PhD candidate and MD/PhD trainee applying mathematical, computational, and machine learning methods to complex biomedical data. Research spans multimodal AI, medical computer vision, genomics, model evaluation, and computational modeling, with software and analyses developed for collaborators across Caltech, USC, and Children's Hospital Los Angeles.

EDUCATION

PhD (Computational Biology), *California Institute of Technology* May 2023 – Apr. 2027

- **Advisor**: Lior Pachter | **Dissertation Title (working)**: Computational Methods and Machine Learning for Radiogenomics

MD, *Keck School of Medicine of the University of Southern California* Aug. 2021 – June 2029

- President of Medical Computer Science Interest Group, Radiology Interest Group, Pathology Interest Group

BS, *University of California, Los Angeles* Sept. 2016 – June 2020

- **Degree**: Molecular, Cell and Developmental Biology; Biomedical Research Minor | **GPA**: 3.99/4.0 | **MCAT**: 524/528 ("100th percentile")

RESEARCH EXPERIENCE

California Institute Of Technology | Multimodal AI & Bioinformatics | Advisor: Lior Pachter, PhD | Sept. 2023 – Present

- Ines
- Developing methods to interpret AI latent spaces, with applications to neurodegenerative disease
- Developing computational and machine-learning methods that connect radiologic imaging with genomic data — RSNA 2026 oral presentation
- Benchmarked single-cell foundation models (scGPT, Geneformer, scFoundation, scVI), measuring how preprocessing choices affect learned embeddings and cell-type predictions — NeurIPS ICBINO-BIO workshop
- Created ctkit, a standardized pipeline for processing CT data for machine learning and demonstrated importance for removing confounding variables in multi-institutional data, with applications to TCGA and CPTAC imaging — Journal of Imaging Informatics in Medicine (pending)
- Co-developed cellsweep, a tool that uses expectation-maximization to estimate and remove noise in single-cell RNA-seq counts — BMC Bioinformatics (pending)
- Built fastQpick, a tool for sampling FASTQ reads with or without replacement to support scalable subsampling and bootstrap analyses — Bioinformatics
- Developed varseek, a pipeline for detecting and analyzing cancer-associated variants in RNA-seq data — Nucleic Acids Research (pending)
- Conducted analysis of single-cell RNA-sequencing data for a prospective clinical trial of CAR T cell therapy in breast cancer — Nature Medicine (pending)
- Compared sources of variability between Seurat and Scanpy in single-cell RNA-seq analyses — Cell Systems

USC Keck School of Medicine | Radiology AI | Advisors: Vinay Duddalwar, MD; Assad Oberai, PhD| Jun 2022 – Present

- Co-developing RADAR, a multimodal, agentic AI assistant combining a large language model, CT image analysis, retrieval-augmented generation, and clinical context for renal mass assessment
- Conducted cost-effectiveness analysis of kidney cancer detection radiology AI models using Markov microsimulation — Urology Practice
- Evaluated deep learning-based renal tumor image imputation models on clinically-relevant questions — PLOS Digital Health
- Developed and evaluated deep learning for CT: bone lesion segmentation with convolutional neural networks and nnU-Net; assessed external validity across institutions. — European Journal of Radiology: Artifical Intelligence

Children's Hospital Los Angeles | Radiology | Advisor: Amit Sura, MD | Feb. 2023 – Present

- Performing multi-institutional retrospective and prospective analysis of a scoliosis AI detection software with The Hospital for Sick Children
- Conducted retrospective clinical studies of radiologist recommendations, chronic lung disease diagnosis, and pediatric imaging outcomes; applied biostatistics and inter-rater agreement analysis.

Teaching EXPERIENCE

MCAT Master Tutor, MedSchoolCoach | Apr 2020–Nov 2025; Private MCAT and Interview Coach, Wyzant | Aug 2021–Present

• Delivered 750+ hours of individualized MCAT instruction to 30+ students; built lesson plans, taught quantitative and science concepts, and guided practice-problem review. Promoted to Master Tutor for strong client ratings.

CERTIFICATIONS, SKILLS & INTERESTS

- **Mathematics & statistics**: probability, statistical modeling, biostatistics, numerical methods, mathematical modeling, optimization, real analysis, information theory, graph theory, dimensionality reduction, microsimulation, reproducibability
- **AI & data science**: artificial intelligence, deep learning, computer vision, convolutional neural networks, vision transformers, foundation models, large language models, multimodal AI, agentic AI, PyTorch, scikit-learn
- **Computing**: Python, R, SQL, Bash, Git, Docker, AWS, REST/GraphQL APIs, web scraping, data visualization, scientific software development
- **Molecular Biology**: single-cell RNA-sequencing, bulk RNA sequencing, whole genome/exome sequencing, spatial transcriptomics, microarray, fluorescence microscopy, western blot, polymerase chain reaction